Geldium | Collections Strategy Brief

Task 3 · Decision memo for the Head of Collections · 16 October 2026

Executive decision

Approve a controlled six-week, service-oriented outreach pilot—not automated collections prioritisation by AI score. In a 500-customer sample, 16% were delinquent. The tested models did not reliably distinguish delinquent from non-delinquent customers on unseen records. Use observable payment-service needs for a modest pilot while improving data quality and validating future models.

1 | Predictive insights: what the evidence actually shows

The dataset contains 80 delinquent customers (16%) and 420 non-delinquent customers. Exploratory differences exist across segments, but these are descriptive associations, not validated causal drivers or reliable prediction rules. The 125-customer holdout contained only 20 delinquent cases.

Top 3 risk factors / monitoring signals (priorities for investigation, not proven drivers):

1. Recent payment behaviour: Monthly payment-status fields (Month_1–Month_6) were among the more influential model inputs in an exploratory permutation analysis (especially Month_4 and Month_1). Their importance is unstable and requires confirmation that records predate the outcome.

2. Repayment capacity: Customers with debt-to-income ratios of 40–50% had an observed delinquency rate of 20.9%, versus 9.8% for ratios of 0–20%. Small subgroup sizes and weak overall correlations prevent a definitive risk claim.

3. Customer context and data quality: Unemployed customers had a 19.4% observed rate versus 11.5% for retired customers; Business card holders had 21.3% versus 11.8% for Platinum. These characteristics must not automatically trigger adverse treatment. Location appeared in exploratory importance rankings and may be a fairness-sensitive proxy.

Model performance and business implications

Holdout evidence	Logistic regression	Random forest	Operational meaning
PR-AUC (baseline 0.160)	0.213	0.244	Modest ranking gains; uncertain with 20 positive cases
ROC-AUC (random = 0.500)	0.457	0.400	Both below random on this test split
At 50% threshold	5/20 found; 44 false alerts	0/20 found; 0 false alerts	Not suitable for automated outreach decisions

Logistic regression ranked highest on training-only cross-validation PR-AUC (0.208); Random Forest led the holdout PR-AUC, but its ROC-AUC was 0.400. These mixed results do not establish a deployable winner. Some previously produced chart labels differ from the saved results CSV; this memo uses the CSV as the numerical reference.

From insight to action

2 | One SMART recommendation

SMART goal: Within 30 days, launch a six-week randomised pilot of supportive SMS/email reminders and optional payment-assistance information for 200 eligible customers with recent, verified late or missed payment events (100 intervention; 100 business-as-usual control), subject to available volume, consent and contact rules. Aim for at least a 10% relative reduction in new 30+ days-past-due cases versus control, while tracking contact cost, opt-outs and complaints. Treat 10% as a pilot target, not a forecast.

Stakeholder rationale: This is operationally feasible using existing outreach channels and directly addresses payment friction without relying on an unvalidated AI score. A randomised comparison will show whether the intervention helps customers and improves collections outcomes before any wider rollout.

Action choices and decision

Option	Feasibility / customer impact	Decision
Supportive reminders + optional assistance	Low-cost, scalable; low intrusion with consent and frequency limits	Pilot first
Financial coaching or tailored payment plans	Potentially helpful; needs trained staff, eligibility and affordability checks	Offer on request / escalation
Automatic limit reductions or aggressive collections	Potential customer harm, bias and compliance risk; no reliable model support	Do not implement

Pilot measurement and decision gates

Primary outcome: New 30+ days-past-due incidence over a consistent follow-up window; compare treatment versus control and report absolute and relative differences with uncertainty intervals. Secondary: Cure rate, payment completion, successful contact, cost per prevented delinquency, complaints, opt-outs and hardship referrals. Monitor weekly; expand only if benefit, customer experience and fairness are acceptable. Six weeks may be too short for complete 30+ DPD observation, so extend follow-up where necessary.

3 | Ethical and responsible AI safeguards

Fairness risk 1 — proxy discrimination: Location, employment and card type may correlate with protected or vulnerable groups. Mitigation: Do not use them for automatic exclusion, punitive actions or differential treatment; audit selection, contact and outcomes by legally appropriate groups, with compliance review.

Fairness risk 2 — unequal model errors and representation: Only 80 positive cases exist overall; subgroup false-positive and false-negative estimates are especially unstable. Mitigation: Report group sizes and confidence intervals, assess disparate impact and error-rate differences when lawful data are available, and suspend any harmful use pending review.

Explainability and oversight: Tell staff that an AI risk score is an uncertain statistical estimate, not proof a customer will miss a payment. Keep reason codes grounded in verified inputs, document data lineage and decisions, provide human review and customer support pathways, and prohibit fully automated adverse credit actions based on this model.

Decision requested and next 30 days

Approve: pilot design, compliance review and measurement ownership. Do not approve: production AI scoring. In parallel, reconcile chart/model versions, validate the timing of payment-history fields, acquire more representative labelled outcomes, and rerun out-of-time validation before reconsidering AI-led targeting.

Evidence base: Geldium 500-row Task 1 dataset; Task 2 model-results CSV and holdout charts. Segment rates are descriptive; no intervention effect or ROI has yet been measured. GenAI assisted with drafting and synthesis; figures and recommendations require human sign-off.